

Dr Wencheng Lau-Medrano

Marine science researcher

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Summary

Ph.D. in Marine Sciences with over eight years of experience in fisheries research, stock assessment, and science-based management. I bring professional experience in Peru and France (IMARPE, TNC, IRD), working across pelagic, demersal, and invertebrate fisheries, and contributing to national and regional scientific processes. My work includes supporting stock assessment processes through the preparation, validation, and integration of fisheries datasets, and the development of transparent, reproducible analytical workflows. I have worked extensively with complex fisheries data, including observer and logbook datasets, from collection and digitisation through to quality control, analysis, and reporting. I am familiar with RFMO scientific processes through participation in international committees (ICCAT, IOTC) and contributions to SPRFMO-related work. I bring strong skills in R, fisheries data systems, and scientific communication, and I am experienced in translating complex analyses into practical outputs that support decision-making. Recognized for delivering high-quality outputs under tight timelines, I work independently and support effective coordination across scientific teams and science-policy interfaces.

Work experience

Researcher and data analyst in Demersal resources and Jumbo squid fishery

TNC, Peru

Jul/2025 - Now

- Conducted exploratory analyses of catch-effort data from national monitoring systems, producing time series and visualizations to support fisheries assessments.
- Developed automated data validation workflows for large fisheries datasets (catch, effort, biological and biometric data), improving data quality and reproducibility.
- Built and implemented analytical tools (R-based) for stock assessment, including automated interfaces for JABBA, comparative analyses with SPiCT, and Applied spatiotemporal modelling (VAST) to standardize CPUE and generate abundance indices for jumbo squid (*Dosidicus gigas*) and Peruvian hake (*Merluccius gayi peruanus*).
- Compiled and structured multi-source fisheries databases, integrating historical and operational datasets into standardized formats.
- Produced technical reports and policy-relevant analyses (e.g., catch limits by fleet segment) to support decision-making by national authorities.
- Contributed to MSE-related work and MSC certification processes, including model sensitivity analyses, documentation, and stakeholder engagement.
- Completed training in international fisheries law, strengthening knowledge on IUU fishing, regulatory frameworks, and RFMO-related governance processes.

Doctoral researcher

IRD, France

Dec/2021 - May/2025

- Conducted doctoral research at the University of Montpellier (MARBEC/IRD) focusing on artificial intelligence applications to understand tropical tuna purse-seine fishery dynamics.
- Analyzed 17 years of vessel trajectories and logbooks to characterize long-term shifts in fishing strategies between free-swimming schools and drifting Fish Aggregating Devices (dFADs).
- Quantified the environmental footprint of deactivated dFADs in the Indian Ocean, obtaining an estimate of their in-water half-life and assessing coastal stranding risks.
- Developed machine learning models, including fully connected neural networks, to predict tuna catch variability using spatio-temporally non-local ocean productivity information.
- Integrated multi-source datasets, such as satellite imagery and vessel monitoring systems, employing statistical tools like Generalized Additive Models and Random Forests.
- Participated in ICCAT and IOTC meetings, attending scientific sessions and presenting results, which provided direct exposure to the science-management interface within these RFMOs.

Researcher and data analyst in Demersal Resources

IMARPE, Peru

Jan/2018 - Oct/2021

- Spatial, statistical and geostatistical analysis in the follow-up and monitoring program of the industrial and artisanal fishery of Peruvian hake.
- Peruvian hake biomass estimation from hydroacoustic information and sampling on board research vessels for annual campaigns between 2018-2021.
- Experience in the elaboration of hake resource status reports.
- Development of R packages (software libraries) for the implementation of biomass estimation methods and automation of information reports from research cruises and from the hake industrial fleet.
- Experience in extraction and calculation of various abundance indices (e.g. standardized CPUE), resource-environment relationships, estimation of fishing selectivity, calculation of growth curves, distribution maps and bycatch estimation within the hake, jumbo squid, blue shark (*Prionace glauca*) and mako (*Isurus oxyrinchus*) fisheries.
- Planning, management and development of a project for IUU (Illegal, Unreported and Unregulated) catch estimation within the Peruvian hake fishery.

Researcher and data analyst in Highly Migratory and Pelagic Resources

IMARPE, Peru

Dec/2013 - Dec/2017

- Spatial, statistical and geostatistical analysis in the PBP ("Programa Bitácoras de Pesca", Fishing On-board Logbook Program), where effort and catch data is collected from industrial vessels dedicated to the extraction of Peruvian anchovy (*Engraulis ringens*).
- Estimation of Peruvian anchovy biomass from hydroacoustic information and sampling on board research vessels through the Simmonds (2009) method for the biannual campaigns between 2014-2017.
- Experience in the elaboration of status reports of the anchoveta resource (north-central stock) and stock evolution from different exploitation scenarios.
- Development of R packages (software libraries) for the implementation of biomass estimation methods and automation of information reports from research surveys and from the industrial fleet.
- Experience in extraction and calculation of several abundance indices (e.g. standardized CPUE), resource-environment relationships, estimation of fishing selectivity, calculation of growth curves, distribution maps and estimation of bycatch within the Peruvian anchoveta, jack mackerel (*Trachurus murphyi*), mackerel (*Scomber japonicus*) and dorado (*Coryphaena hippurus*) fisheries.
- Experience in the development and execution of several courses: R (basic, intermediate and advanced levels), GIS, Shiny, R markdown, Quarto, and LaTeX.
- Participation as speaker in several national (Peru) and international congresses.

Publications

Trends and drivers of French tropical tuna purse-seine vessel behavior in the Atlantic and Indian Oceans, 2004-2020

Fisheries Research

May, 2025

- Wencheng Lau-Medrano, Daniel Gaertner, Francis Marsac and David M. Kaplan
- <https://doi.org/10.1016/j.fishres.2025.107330>

First look at the distribution of deactivated dFADs used by the French Indian Ocean tropical tuna purse-seine fishery

ICES Journal of Marine Science

Jul, 2024

- Wencheng Lau-Medrano, Daniel Gaertner, Francis Marsac, Lorelei Guéry and David M. Kaplan
- <https://doi.org/10.1093/icesjms/fsae104>

Spatio-temporal distribution modelling of dolphinfish in the Pacific Ocean off Peru using artisanal longline fishery data

Deep Sea Research Part II: Topical Studies in Oceanography

Oct, 2019

- Josymar Torrejón-Magallanes, Daniel Grados and Wencheng Lau-Medrano
- <https://doi.org/10.1016/j.dsr2.2019.104665>

Education

University of Montpellier

Doctor in Marine Science

Montpellier, France

Dec 2021 - May 2025

- Project name: Artificial intelligence for a better understanding of stock size and fishery dynamics in the Indian Ocean purse seine tropical tuna fishery.
- Advisor: David M. Kaplan

Universidad Peruana Cayetano Heredia

Master in Marine Science

Lima, Peru

Apr 2014 - Jan 2017

- Thesis title: Construction of biomass indices for the anchoveta (*Engraulis ringens*) and jack mackerel (*Trachurus murphyi*) resources of the Peruvian Upwelling Ecosystem using empirical models based on acoustic information.
- Advisor: Ricardo Oliveros-Ramos

Universidad Nacional Mayor de San Marcos

Bachelor in Biology

Lima, Peru

Apr 2007 - Sep 2012

- Orientation in "Hydrobiology and Fisheries".

Internship experience

University of Montpellier - Marbec

Contributing to the development and exploration of spatially and environmentally explicit operational models for jumbo squid assessment within the SPRFMO framework.

Montpellier, France

Mar-Apr, 2026

University of Montpellier - Marbec

Performing a calibration of the OSMOSE model for the Peruvian Upwelling System

Montpellier, France

Feb-Apr, 2020

University of Washington - SAFS

Performing a CPUE standardization using spatial-temporal approach (VAST model)

Seattle, USA

Sept-Oct, 2018

Software

imagine: IMAGInG engines, Tools for Application of Image Filters to Data Matrices

Provides fast application of image filters to data matrices, using R and C++ algorithms.

R, C++

grec: Gradient-Based Recognition of Spatial Patterns in Environmental Data

Provides algorithms for detection of spatial patterns from oceanographic data using image processing methods based on Gradient Recognition.

R

Skills

Participacion in research surveys

At IMARPE, I have participated in pelagic (midwater trawl) and demersal (bottom trawl) research surveys, including fishing operations, biological sample collection, and laboratory/field data analysis (2016–2021). Additionally, participated in krill resource assessment surveys in the Bransfield Strait and near King George Island, Antarctica (2014, 2017).

Programming

Advanced R, Quarto and Shiny, intermediate C++ and Python. Experience with the use of Machine Learning and Deep learning modeling with torch (R) and Pytorch.

Languages

Spanish (mother tongue), English (high), French (basic).

Soft-skills

Critical and lateral thinking, problem-solving, creativity and innovation, adaptability, resilience, patience, strong ability to work independently, and rapid learning capacity.